

Meissner Minimality in the Confocal Peabody Family

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September 30, 2026

Abstract

Arelio, Montejano, and Oliveros introduced a three-parameter family of bodies of constant width obtained from a regular Reuleaux tetrahedron by replacing its six edge neighborhoods with envelopes of spheres associated with confocal quadrics. The family contains the two classical Meissner bodies as a circle-line degeneration and Robert’s tetrahedrally symmetric body as a parabolic limit. We prove that the Meissner degenerations minimize volume within this regular-tetrahedron confocal family. The geometric part of the proof gives the exact decomposition

$$\text{Vol } K(\mathbf{e}, \boldsymbol{\sigma}) = V_M + \Phi(e_1) + \Phi(e_2) + \Phi(e_3),$$

independently of the eight orientation choices. A canonical confocal chart reduces the one-pair contribution to a fixed one-dimensional integral. A finite outward-rounded interval computation proves $\Phi(e) > 0$ for every $0 < e < 1$, using 2,590 terminal boxes. Equality holds exactly when all three parameters vanish. The result is restricted to regular-tetrahedron confocal Peabodies and does not resolve the three-dimensional Blaschke–Lebesgue problem.

Keywords: bodies of constant width, Peabodies, Meissner bodies, confocal quadrics, interval arithmetic, computer-assisted proof

1 Introduction

The Blaschke–Lebesgue problem asks for convex bodies of minimum volume among all convex bodies of prescribed constant width. In the plane, the minimizers are the Reuleaux triangles [3, 7]. The corresponding three-dimensional problem remains open. Bonnesen and Fenchel conjectured that the two noncongruent Meissner bodies are minimizers; their common width-one volume is

$$V_M = \pi \left(\frac{2}{3} - \frac{\sqrt{3}}{4} \arccos \frac{1}{3} \right) = 0.419860045965080 \dots \quad (1.1)$$

See [6, 8, 10] for background.

Arelio, Montejano, and Oliveros introduced *Peabodies*, obtained by replacing neighborhoods of the six edges of a Reuleaux tetrahedron with sections of envelopes of spheres guided by confocal quadrics [1]. Their construction gives genuine bodies of constant width and contains both Meissner bodies and Robert’s tetrahedrally symmetric body. We restrict to the family based on a regular tetrahedron.

The regular tetrahedron has three pairs of opposite edges. Each pair has a parameter $e_i \in [0, 1)$, and an orientation bit $\sigma_i \in \{0, 1\}$ records which edge receives the elliptic device. Let

$$\mathbf{e} = (e_1, e_2, e_3), \quad \boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3),$$

*Pre-circulation draft. The independent formula and release audit should be completed before submission.

and write $K_d(\mathbf{e}, \boldsymbol{\sigma})$ for the resulting body of width d . The value $e_i = 0$ is the circle-line Meissner degeneration; the limit $e_i \rightarrow 1^-$ is parabolic.

Our main result is the following.

Theorem 1.1 (Main theorem). *For every $d > 0$, every $\mathbf{e} \in [0, 1]^3$, and every $\boldsymbol{\sigma} \in \{0, 1\}^3$,*

$$\text{Vol } K_d(\mathbf{e}, \boldsymbol{\sigma}) \geq \pi \left(\frac{2}{3} - \frac{\sqrt{3}}{4} \arccos \frac{1}{3} \right) d^3. \quad (1.2)$$

Equality holds if and only if $e_1 = e_2 = e_3 = 0$. Within this family, the equality cases are the two classical Meissner bodies.

The proof has two parts. First, a partition of the unit normal sphere gives an exact additive volume formula, reducing the three-parameter problem to one scalar function Φ . Second, a finite interval certificate proves $\Phi(e) > 0$ for $0 < e < 1$. The computer-assisted component is narrow: all formulas and reductions are proved analytically, while the remaining sign verification is reduced to finitely many outward-rounded interval inequalities.

The paper is organized as follows. Section 2 recalls the Peabody construction and proves the additive volume formula. Section 3 derives the one-pair integral. Section 4 proves the scalar positivity theorem. Section 5 proves Theorem 1.1. Section 6 records the scope and reproducibility of the argument. The stable chart and interval details are collected in the appendices.

2 Peabody geometry and exact additivity

We begin at the native scale of the Peabody construction: a regular tetrahedron of side length two and a body of constant width two.

2.1 Confocal devices and the source construction

A pea-pod device is a family of balls whose centers move along a conic and whose radii vary so that the balls remain tangent to two fixed circles. Two devices are confocal when their planes are orthogonal, their axes coincide, and their center curves are confocal quadrics. The basic identity proved in [1] is

$$|x - y| + R(x) + R(y) = L, \quad (2.1)$$

where x, y lie on the two center curves and L is independent of x, y . For opposite edges of a regular tetrahedron of side two, $L = 2$.

The two endpoints on the line through x and y at distances $R(x)$ and $R(y)$ sweep out a pair of wedge-pod surfaces. Their joining segment has length two and is normal to both surfaces. Applying this construction to all three pairs of opposite edges gives six wedge-pod surfaces; four spherical caps close the remaining vertex regions. Arelio, Montejano, and Oliveros prove that these ten patches form the boundary of a convex body of constant width two [1].

2.2 The normal-sphere decomposition

For a strictly convex body K of constant width two, let $R_K(n)$ be the support point with outward normal $n \in \mathbb{S}^2$.

Lemma 2.1. *For every $n \in \mathbb{S}^2$,*

$$R_K(-n) = R_K(n) - 2n. \quad (2.2)$$

Proof. Put $x = R_K(n)$ and $y = R_K(-n)$. The supporting planes have separation two, so $(x - y) \cdot n = 2$. Since a body of constant width two has diameter two, $|x - y| \leq 2$. Equality in Cauchy–Schwarz gives $x - y = 2n$. $\square$

Let C_A, C_B, C_C, C_D be the four radius-two spherical caps and let $\Omega_A, \dots, \Omega_D \subset \mathbb{S}^2$ be their Gauss images. By Theorem 2.1, the normal cone at a vertex is the antipodal image of the Gauss image of the opposite cap. For the i -th opposite-edge pair, write the two wedge-pod surfaces as W_i^+, W_i^- , and put

$$\Gamma_i = \text{Gauss}(W_i^+).$$

The source binormal pairing gives

$$\text{Gauss}(W_i^-) = -\Gamma_i.$$

Ignoring seams of spherical measure zero, the smooth Gauss images and the four vertex normal cones partition $\mathbb{S}^2$. Hence

$$4\pi = 2 \sum_{v \in \{A, B, C, D\}} |\Omega_v| + 2 \sum_{i=1}^3 |\Gamma_i|. \quad (2.3)$$

A patch on a sphere of radius two has area four times the area of its Gauss image. Therefore the total cap area is

$$\sum_v |C_v| = 8\pi - 4 \sum_{i=1}^3 |\Gamma_i|. \quad (2.4)$$

Adding the wedge-pod areas gives

$$S_2 = 8\pi + \sum_{i=1}^3 \Psi(e_i), \quad \Psi(e_i) := |W_i^+| + |W_i^-| - 4|\Gamma_i|. \quad (2.5)$$

Exchanging the elliptic and hyperbolic devices merely exchanges W_i^+ and W_i^- , so Ψ is independent of the orientation bit.

For a body of constant width d , Blaschke's identity is [5, 2, 4]

$$V = \frac{d}{2}S - \frac{\pi}{3}d^3. \quad (2.6)$$

At width two, (2.5) gives

$$V_2 = \frac{16\pi}{3} + \sum_{i=1}^3 \Psi(e_i).$$

Scaling by one half multiplies volume by 1/8. Define

$$\Phi(e) := \frac{\Psi(e) - \Psi(0)}{8}. \quad (2.7)$$

Since the all-zero body is Meissner, we obtain the exact reduction

Proposition 2.2. *For every width-one regular-tetrahedron confocal Peabody,*

$$\text{Vol } K_1(\mathbf{e}, \boldsymbol{\sigma}) = V_M + \Phi(e_1) + \Phi(e_2) + \Phi(e_3), \quad (2.8)$$

independently of $\boldsymbol{\sigma}$.

3 The one-pair formula

Fix one opposite-edge pair and an orthonormal frame $(\mathbf{k}, \mathbf{p}, \mathbf{q})$. Let $e \in [0, 1)$ and set

$$b^2 = \frac{3 + 3e^2 + 4\sqrt{2}e}{1 - e^2}, \quad a = \frac{b}{\sqrt{1 - e^2}}. \quad (3.1)$$

Define

$$\theta = \arccos \frac{1}{b}, \quad \eta = \operatorname{arsinh} \frac{1}{b}, \quad u_0 = \sin \theta, \quad v_0 = \cosh \eta.$$

The two center curves are

$$X(t) = a \sin t \mathbf{k} + b \cos t \mathbf{p}, \quad \theta \leq t \leq \pi - \theta, \quad (3.2)$$

$$Y(\xi) = ae \cosh \xi \mathbf{k} + b \sinh \xi \mathbf{q}, \quad -\eta \leq \xi \leq \eta. \quad (3.3)$$

A direct calculation gives

$$d := |X - Y| = a(\cosh \xi - e \sin t), \quad (3.4)$$

while the pea radii are

$$R_E = ae(\sin t - u_0), \quad R_H = a(v_0 - \cosh \xi). \quad (3.5)$$

The regular beam relation is $a(v_0 - eu_0) = 2$, and therefore

$$d + R_E + R_H = 2. \quad (3.6)$$

Let $n = (X - Y)/d$. Differentiation gives

$$n_t \cdot n_\xi = 0, \quad |n_t| = |n_\xi| = \frac{b}{d}. \quad (3.7)$$

Thus the spherical area element of the Gauss image is

$$d\omega = \frac{b^2}{d^2} dt d\xi. \quad (3.8)$$

The paired surface points are $U_E = X + R_E n$ and $U_H = Y - R_H n$. Using (3.6),

$$\begin{aligned} (U_E)_t &= (2 - R_H)n_t, & (U_E)_\xi &= R_E n_\xi, \\ (U_H)_t &= -R_H n_t, & (U_H)_\xi &= -(2 - R_E)n_\xi. \end{aligned}$$

Substitution into the definition of Ψ yields

Proposition 3.1. *For $0 \leq e < 1$,*

$$\Psi(e) = -2b^2 \int_\theta^{\pi-\theta} \int_{-\eta}^\eta \frac{d + R_E R_H}{d^2} d\xi dt. \quad (3.9)$$

The ξ -integration is elementary. Put

$$I(C) = \frac{4}{\sqrt{1 - C^2}} \arctan \left(\tanh \frac{\eta}{2} \sqrt{\frac{1 + C}{1 - C}} \right). \quad (3.10)$$

Then

$$\begin{aligned} \Psi(e) = -4b^2 \int_{\theta}^{\pi/2} & \left[\frac{I(e \sin t)}{a} \right. \\ & \left. + \frac{e(\sin t - u_0)}{1 - e^2 \sin^2 t} \left((v_0 e \sin t - 1)I(e \sin t) + \frac{2}{b} \right) \right] dt. \end{aligned} \quad (3.11)$$

At $e = 0$,

$$\Psi(0) = -\frac{2\pi}{\sqrt{3}} \arccos \frac{1}{3}, \quad (3.12)$$

which recovers (1.1) from (2.8). The substitution $x = b \cos t$ gives a fixed-interval formula

$$\Psi(e) = \int_0^1 H(e, x) dx. \quad (3.13)$$

For the computer-assisted proof we use the stable parameter

$$q = \frac{1-e}{1+e}, \quad e = \frac{1-q}{1+q}. \quad (3.14)$$

The explicit formula for $H(q, x)$ and its branch control are given in Appendix A.

4 Certified positivity of the scalar increment

The remaining task is

$$\Phi(e) > 0 \quad (0 < e < 1). \quad (4.1)$$

The proof does not require global monotonicity of Φ , nor pointwise positivity of the integrand in (3.13).

At the Meissner endpoint, exact differentiation of (3.11) gives

$$\begin{aligned} \Phi'(0) = \frac{1}{2} & \left[\frac{5\pi}{3\sqrt{3}} - 3 \right. \\ & \left. + \arccos \frac{1}{3} \left(\frac{3\sqrt{2}}{2} - \frac{5\sqrt{6}\pi}{18} \right) \right] = 0.0014901042070382105 \dots > 0. \end{aligned} \quad (4.2)$$

The parameter interval is split exactly into

$$\left[0, \frac{1}{100} \right], \quad \left[\frac{1}{100}, \frac{199}{200} \right], \quad \left[\frac{199}{200}, 1 \right).$$

The following proposition is verified by outward-rounded interval arithmetic; Appendix B describes the finite reductions.

Proposition 4.1 (Scalar certificate). *The function Φ satisfies:*

$$\frac{\Phi(e)}{e} \geq \frac{1}{2000}, \quad 0 < e \leq \frac{1}{100}, \quad (4.3)$$

$$\Phi(e) > \frac{1}{10^7}, \quad \frac{1}{100} \leq e \leq \frac{199}{200}, \quad (4.4)$$

$$\Phi(e) > \frac{1}{20000}, \quad \frac{199}{200} \leq e < 1. \quad (4.5)$$

Table 1: Summary of the interval certificates.

Block	Range	Certified bound	Weakest raw lower endpoint	Boxes
Local	$0 < e \leq 1/100$	$\Phi(e)/e \geq 1/2000$	$5.4158529808 \cdot 10^{-4}$ for Φ'	160
Bulk	$1/100 \leq e \leq 199/200$	$\Phi(e) > 10^{-7}$	$3.2899516604 \cdot 10^{-7}$	430
Tail	$199/200 \leq e < 1$	$\Phi(e) > 1/20000$	$6.0546375326 \cdot 10^{-5}$	2,000

The local certificate uses 160 terminal boxes, the bulk certificate 430, and the tail certificate 2,000. All parameter endpoints are exact rationals. The two seams agree exactly, since

$$e = \frac{199}{200} \iff q = \frac{1}{399}.$$

The weakest raw lower bounds are shown in Table 1.

Theorem 4.2. *One has*

$$\Phi(0) = 0, \quad \Phi(e) > 0 \quad (0 < e < 1). \quad (4.6)$$

Proof. The identity $\Phi(0) = 0$ follows from (2.7). The three parts of Theorem 4.1 cover every $0 < e < 1$, with exact overlap at the two seams. $\square$

5 Proof of the main theorem

Proof of Theorem 1.1. By dilation it is enough to take $d = 1$. Proposition 2.2 and Theorem 4.2 give

$$\text{Vol } K_1(\mathbf{e}, \boldsymbol{\sigma}) - V_M = \Phi(e_1) + \Phi(e_2) + \Phi(e_3) \geq 0.$$

If equality holds, every summand vanishes, so $e_1 = e_2 = e_3 = 0$. Conversely, the all-zero triple is the circle-line degeneration on every opposite-edge pair and gives a classical Meissner body. Scaling by d proves (1.2). $\square$

6 Discussion and reproducibility

The theorem proves Meissner minimality only in the regular-tetrahedron confocal Peabody family. It does not prove the global Meissner conjecture, improve a universal lower bound, or treat arbitrary Peabodies arising from other ball polyhedra. In particular, the pairwise semi-regular construction discussed in [1] does not by itself provide a global six-patch constant-width assembly for a deformed tetrahedral base.

The exact part of the proof consists of the source Peabody construction, the normal-sphere decomposition, the width normalization, and the one-pair formulas. The finite verification uses exact rational partition endpoints and `mpmath 1.3.0` interval arithmetic at 80 decimal digits. The central certificate was replayed at 100 digits and in reverse subdivision order; the tail certificate was replayed at 60, 80, and 100 digits. Deliberate sign and baseline mutations were detected, and an under-resolved partition was rejected. The complete proof uses 2,590 terminal interval boxes.

The frozen supplementary archive [11] has SHA-256

b07df7201d2c5dcfaee65fa90c85a643f2bd55e81ce56fbf924b20e04e632459.

It contains the exact formulas, all terminal boxes, machine-readable certificates, adversarial audits, and deterministic replay scripts. The computer-assisted statement is therefore made under the documented trust boundary of the Python interpreter, `mpmath` interval implementation, and the underlying arithmetic libraries.

A The stable fixed-interval chart

Put

$$\kappa = 1 + \sqrt{2}, \quad K = \kappa^2 = 3 + 2\sqrt{2}.$$

For $0 \leq q \leq 1$ and $0 \leq x \leq 1$, define

$$\begin{aligned} D &= \sqrt{K^2 + q^2}, \\ R &= \sqrt{K^2 + q^2 - 2Kqx^2}, \\ T &= \sqrt{2K(K^2 + q^2) + K^2(1 - q)^2x^2}, \\ \delta &= \frac{2K(1 - x^2)}{R + K - q}, \\ M &= \frac{K(q - 2Kx^2)}{R + K} + (1 - K)R - K^2 - qR - q - q^2, \end{aligned}$$

and

$$\begin{aligned} P_1 &= -\frac{16\sqrt{2}\kappa(K^2 + q^2)}{RT}, \\ P_2 &= -\frac{4K(1 - q^2)(K^2 + q^2)\delta M}{RT^3}, \\ Q_0 &= -\frac{4K(1 - q^2)(K^2 + q^2)\delta}{RT^2}, \\ Z &= \frac{K((1 + q)D + (1 - q)R)}{T(K + q + D)}. \end{aligned}$$

Then

$$H(q, x) = (P_1 + P_2) \arctan Z + Q_0, \quad \Psi(e(q)) = \int_0^1 H(q, x) \, dx. \quad (\text{A.1})$$

The formula is obtained from (3.11) by $x = b \cos t$, followed by the substitution (3.14) and rationalization of the removable differences near $q = 0$. The useful identities are

$$1 - e^2 = \frac{4q}{(1 + q)^2}, \quad b^2 = \frac{K^2 + q^2}{2Kq}, \quad a^2 = \frac{(K^2 + q^2)(1 + q)^2}{8Kq^2}.$$

Moreover,

$$R^2 \geq (K - q)^2 > 0, \quad T^2 > 0, \quad R + K - q > 0, \quad K + q + D > 0.$$

Thus all radicals use the positive real branch, all denominators are nonzero, and (A.1) extends continuously to $q = 0$.

B Finite interval verification

For interval arithmetic background, see [9]. For an x -panel $[a, b]$ of width h and midpoint m , the central certifier uses

$$\int_a^b f(x) \, dx \in hf(m) + \frac{h^3}{24} f''([a, b]). \quad (\text{B.1})$$

The midpoint Peano kernel is nonnegative and has total mass $h^3/24$. A nested interval automatic-differentiation jet evaluates

$$H, \quad H_q, \quad H_{xx}, \quad H_{qxx}.$$

For a parameter slab $Q = [q_-, q_+]$ with midpoint q_m , the mean-value enclosure

$$\Psi(Q) \subseteq \Psi(q_m) + (Q - q_m)\Psi_q(Q) \quad (\text{B.2})$$

controls the entire slab. Subtracting an independent enclosure of $\Psi(0)$ and dividing by eight gives an enclosure for Φ .

Near $e = 0$, the certificate encloses the integrated derivative and proves $\Phi'(e) > 1/2000$ on $[0, 1/100]$; integration gives (4.3). On the compact bulk, 43 rational parameter slabs and 10 x -panels per slab prove (4.4). In the tail, the stable formula extends to $q = 0$, and five rational q -slabs with 400 x -panels each prove (4.5) directly.

All accepted intervals have strictly positive lower endpoints. The complete manifest records exact endpoints, precision, software versions, terminal-box data, and SHA-256 hashes. Ordinary high-precision quadrature is used only as audit evidence, not as proof authority.

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